

# IPL DATA ANALYSIS

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## An End-to-End Analytics Project using Power BI

*Analysing a decade of Indian Premier League cricket (2008–2017)*

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Tools: Power BI Desktop · Power Query · DAX

Dataset: 636 matches · 150,460 deliveries

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# 1. Introduction

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The Indian Premier League (IPL) is one of the most-watched cricket tournaments in the world, and every season it produces an enormous amount of data. Behind the boundaries and wickets that fans see on screen sits a detailed record of every match played and every single ball bowled. This project takes that raw data and turns it into something anyone can understand at a glance: a set of clear, interactive dashboards that answer real questions about the game.

I worked with two datasets covering ten IPL seasons, from 2008 to 2017. Together they describe 636 matches and over 150,000 deliveries. My goal was to move through the full analytics workflow the way a data analyst does on the job — starting from messy raw files, cleaning and modelling them, writing the calculations, and finally building a dashboard that tells the story clearly.

## 1.1 Objectives

The project set out to answer a specific set of questions about teams, players, and match conditions. In particular, I wanted the final dashboard to be able to show:

- The most successful teams, and how often the team that wins the toss goes on to win the match.
- The highest and lowest run scorers, and who hit the most fours and sixes — both as individual players and as teams.
- The leading wicket-takers, the bowlers who never took a wicket, and the detailed stats of the very best bowlers.
- The best fielders, measured by their catches and run-outs.
- How toss decisions changed over time, which cities hosted the most matches, and the venue where the top team wins most often.

## 1.2 Why this project matters

Beyond the cricket itself, this project is a complete demonstration of the analytics process: taking data that is not ready to use, making it reliable, and communicating what it means. It brought together data cleaning in Power Query, data modelling, calculations in DAX, and dashboard design — the same skills used to analyse sales, operations, or customer data in a business setting.

## 2. About the Dataset

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The analysis is built on two files that work together like two connected tables in a database. One describes each match at a summary level, and the other records every individual ball.

### 2.1 The matches file

This file holds one row per match — 636 rows in total. Each row summarises a complete game: who played, who won the toss and what they chose to do, who won the match and by how much, and where it was played.

| Column                       | What it describes                                                    |
|------------------------------|----------------------------------------------------------------------|
| id                           | A unique number for each match (used to link to the deliveries file) |
| season                       | The year the match was played (2008 to 2017)                         |
| city / venue                 | Where the match took place                                           |
| team1 / team2                | The two teams playing                                                |
| toss_winner / toss_decision  | Who won the toss and whether they chose to bat or field              |
| winner                       | The team that won the match                                          |
| win_by_runs / win_by_wickets | The margin of victory                                                |
| player_of_match              | The best player of the match                                         |

### 2.2 The deliveries file

This is the detailed file — one row for every ball bowled, which is why it contains over 150,000 rows. Every delivery records who bowled, who batted, how many runs were scored, and whether a wicket fell.

| Column                                              | What it describes                                     |
|-----------------------------------------------------|-------------------------------------------------------|
| match_id                                            | Links each ball back to its match in the matches file |
| inning / over / ball                                | Where in the game the delivery happened               |
| batting_team / bowling_team                         | The teams involved on this ball                       |
| batsman / bowler / non_striker                      | The players involved                                  |
| batsman_runs / total_runs                           | Runs off the bat, and total runs including extras     |
| wide_runs / bye_runs /<br>legbye_runs / noball_runs | The different types of extras                         |
| player_dismissed / dismissal_kind<br>/ fielder      | Who got out, how, and which fielder was involved      |

### 2.3 How the two files connect

The two files are joined through a single link: the **id** column in the matches file matches the **match\_id** column in the deliveries file. Because one match contains hundreds of deliveries, this is a one-to-many relationship. This link is what allows a filter on the matches side — say, picking a season — to instantly update every ball-by-ball chart on the other side.

### 3. Tools and Methodology

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I built the entire project in Power BI Desktop, using its different components at each stage. The approach followed a clear sequence, and each tool had a specific job.

| Tool                    | What I used it for                                                                                  |
|-------------------------|-----------------------------------------------------------------------------------------------------|
| Power Query             | Cleaning the raw data — fixing inconsistent names, filling blanks, and setting correct data types   |
| Data Model (Model view) | Connecting the two tables and building a small helper table for teams                               |
| DAX                     | Writing all the calculations — wins, strike rates, economy rates, wickets, fielding stats, and more |
| Power BI report canvas  | Designing the three interactive dashboard pages with slicers and cross-filtering                    |

The methodology moved through five stages, in order: first understanding the raw data, then cleaning it, then modelling it, then writing the measures, and finally designing the dashboard. Working in this sequence meant that each stage rested on solid foundations — there was no point writing calculations before the data was clean, or designing charts before the calculations existed.

### 4. Data Cleaning

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Raw data is almost never ready to analyse straight away, and this dataset was no exception. All cleaning was done in Power Query, which records each step so the process is repeatable and the original files stay untouched. These were the main issues I found and fixed:

- **Inconsistent team names.** One franchise appeared under two spellings — 'Rising Pune Supergiant' and 'Rising Pune Supergiants'. Left as-is, this would have split one team into two. I replaced the plural version with the singular across all six columns that hold team names.
- **Matches with no winner.** Three matches had a blank winner because they were tied or abandoned. I labelled these as 'No Result' so they would never be miscounted as a win for any team.
- **Missing cities.** Seven matches had no city recorded, and all of them were played at the Dubai venue. I filled these in with 'Dubai'.
- **Duplicate venue names.** One stadium in Mohali was listed under two slightly different names, which would have split its match count. I merged them into a single name.
- **Wrong data types.** Several columns, including season and the win-margin columns, were not set as numbers. I corrected the data types so that sorting, summing, and date logic would all work correctly.

One decision worth noting: the **player\_dismissed**, **dismissal\_kind**, and **fielder** columns are blank on most rows. This is not missing data — it simply means no wicket fell on that ball, which is true for about 95% of deliveries. So I deliberately left those blanks in place rather than filling them, because a blank correctly represents 'no dismissal here'.

## 5. Data Model and Calculations

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### 5.1 The model

After cleaning, I connected the two tables in Model view by linking matches[id] to deliveries[match\_id] as a one-to-many relationship. I also created a small **Teams** table — a clean list of every unique team — to drive a single team filter that works across the whole report. Keeping all my calculations in a dedicated Measures table kept the model tidy and easy to navigate.

### 5.2 The calculations (DAX measures)

With the model in place, I wrote a set of measures to calculate everything the dashboard needed. Some of the key ones included:

| Measure                                  | What it calculates                                     |
|------------------------------------------|--------------------------------------------------------|
| Total Wins / Win %                       | How many matches each team won, and their win rate     |
| Batsman Runs / Strike Rate               | Runs scored and scoring speed for each batsman         |
| Bowler Wickets / Economy Rate            | Wickets credited to the bowler and their runs per over |
| Catches / Run Outs / Fielding Dismissals | Each fielder's contribution in the field               |
| Toss to Match Win %                      | How often the toss winner also won the match           |

Writing these correctly required attention to real cricket rules. For example, run-outs are not credited to the bowler, so they were excluded from the bowler's wickets. Similarly, wides and no-balls are not legal deliveries, and byes and leg-byes are not charged to the bowler — all of which had to be handled in the formulas so the numbers matched how cricket is actually scored.

## 6. Analysis and Findings

The dashboard is organised into three pages, each focusing on a different aspect of the game. Every page has slicers so the viewer can filter by season, team, and player, and clicking any chart filters the rest of the page. The following sections walk through each page and what the data revealed.

### 6.1 Page 1 — Teams, Toss and Venues

This page looks at team success and the role of the toss. The four cards at the top give the headline totals: 636 matches across 10 seasons, and a toss-to-match win rate of just over 51%.



Figure 1: The Teams, Toss & Venues dashboard page.

The findings from this page were some of the most interesting in the whole project:

- **Mumbai Indians are the most successful team**, with 92 wins, followed by Chennai Super Kings on 79. Looking at win rate rather than raw totals, Chennai actually come out on top, winning around 60% of their matches.
- **The toss barely matters.** The team that won the toss went on to win the match only about 51% of the time — almost the same as a coin flip. This is a nice example of data challenging a common assumption, since fans often treat the toss as a big advantage.
- **Teams strongly prefer to field first.** Of all the toss decisions, 363 chose to field and only 273 chose to bat — a clear preference for chasing. This pattern also holds at team level: when the tosses-won chart is broken down by decision, almost every team shows a strong lean towards fielding first rather than batting.
- **Mumbai Indians also win the most tosses (85)**, followed by Kolkata Knight Riders (78) and Delhi Daredevils (72) — broadly in line with how many matches each side has played over the ten seasons.

- **Winning the toss does not guarantee winning the match.** Mumbai Indians converted their toss wins into match wins most often (48 times), ahead of Kolkata (44) and Chennai (42), but these numbers are far below their total toss wins — reinforcing that the toss is not the decisive factor fans often assume.
- **Mumbai's home ground is their lucky stadium.** When the dashboard is filtered to Mumbai Indians, the Wankhede Stadium stands out as the venue where they win most often, which fits with it being their home ground.
- **Mumbai and Bangalore host the most cricket.** Looking at matches by city, Mumbai staged the most games (85), followed by Bangalore, Kolkata, and Delhi — the traditional big-city venues that anchor the tournament each season.

## 6.2 Page 2 — Batting

This page focuses on run-scoring and boundary-hitting. The cards show the overall scale of batting across the decade: over 194,000 runs, more than 17,000 fours, and 6,523 sixes.



Figure 2: The Batting dashboard page.

Splitting the boundary charts into separate 'fours' and 'sixes' views made the story much clearer, and a few names stood out:

- **Suresh Raina is the leading run scorer** with 4,548 runs, just ahead of Virat Kohli on 4,423 and Rohit Sharma on 4,207.
- **Chris Gayle is the undisputed six-hitting king**, with 266 sixes — far clear of anyone else. This was a case where sorting mattered: ranking by fours alone would have hidden Gayle's dominance, so a dedicated sixes chart was important.
- **Gautam Gambhir hit the most fours**, with 484, showing a different style of scoring built on placement rather than power.



- **At team level, Mumbai Indians hit the most fours (2,145)**, while Royal Challengers Bangalore hit the most sixes (935) — reflecting their reputation for powerful batting line-ups.
- **At the other end, the lowest scorers** (among batsmen who faced a fair number of balls) include names such as SK Warne with 198 runs, AA Jhunjunwala with 217, and K Goel with 218 — mostly bowlers or lower-order players who batted regularly but were not there to score big.

For the lowest scorers, I added a filter requiring at least 200 balls faced. Without it, the list would have been full of players who faced only a ball or two, which would not be a meaningful comparison. The filter makes sure it shows genuinely low-output batsmen among players who actually spent time at the crease.

### 6.3 Page 3 — Bowling and Fielding

The final page covers the bowlers and fielders. The cards show 145,267 balls bowled and 7,438 total wickets across the ten seasons.



Figure 3: The Bowling & Fielding dashboard page.

This page brought together several of the brief's requirements in one place:

- **Lasith Malinga is the leading wicket-taker** with 154 wickets, ahead of Amit Mishra (134) and Harbhajan Singh (127).
- **The top-five bowler table** shows their full stat lines side by side. Malinga stands out not just for wickets but for his economy of 6.88 — taking the most wickets while also being hard to score off.
- **Around 46 bowlers bowled at least one legal ball but never took a wicket**, mostly part-time or occasional bowlers. The zero-wicket table captures this opposite end of the spectrum, exactly as the brief asked.
- **Dinesh Karthik and MS Dhoni lead the fielding**, with 127 and 126 dismissals respectively — unsurprising, since both are wicket-keepers and so are involved in a large share of catches and stumpings.

## 7. Key Insights

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Pulling the three pages together, a few clear messages emerge from ten seasons of IPL data:

- **Consistency beats everything.** Mumbai Indians and Chennai Super Kings sit at the top not because of one great season but because of sustained success across the decade.
- **The toss is overrated.** At a 51% conversion rate, winning the toss gives almost no real advantage — a genuinely surprising finding for most fans.
- **Chasing is preferred, and slightly favoured.** Teams overwhelmingly choose to field first, and the data shows the chasing side does tend to come out ahead.
- **Star players define eras.** Gayle's sixes, Malinga's wickets, and Raina's runs are the kind of individual dominance that shapes how the tournament is remembered.
- **Wicket-keepers dominate fielding stats,** which is expected given how many dismissals pass through their hands.

## 8. Conclusion and Future Scope

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This project took raw, ball-by-ball IPL data and turned it into a clear, interactive product that answers real questions about the tournament. Along the way it covered the full analytics workflow: understanding the data, cleaning it in Power Query, modelling the relationships, writing calculations in DAX, and designing a three-page dashboard that lets anyone explore the findings for themselves.

The biggest lesson was how much of the value comes before any chart is drawn. Getting the team names consistent, handling the no-result matches, and encoding the correct cricket rules into the measures were what made the final numbers trustworthy. A polished dashboard built on messy data would simply have produced confident-looking but wrong answers.

There is plenty of room to take the project further. Some ideas for future work include:

- Adding player-versus-player analysis — for example, how a specific batsman performs against a specific bowler.
- Extending the dataset to include the most recent seasons, so the trends stay current.
- Building a simple prediction model that estimates a team's chance of winning based on the toss, venue, and recent form.
- Publishing the dashboard online so it can be shared with a single link and refreshed automatically.

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*End of Report*